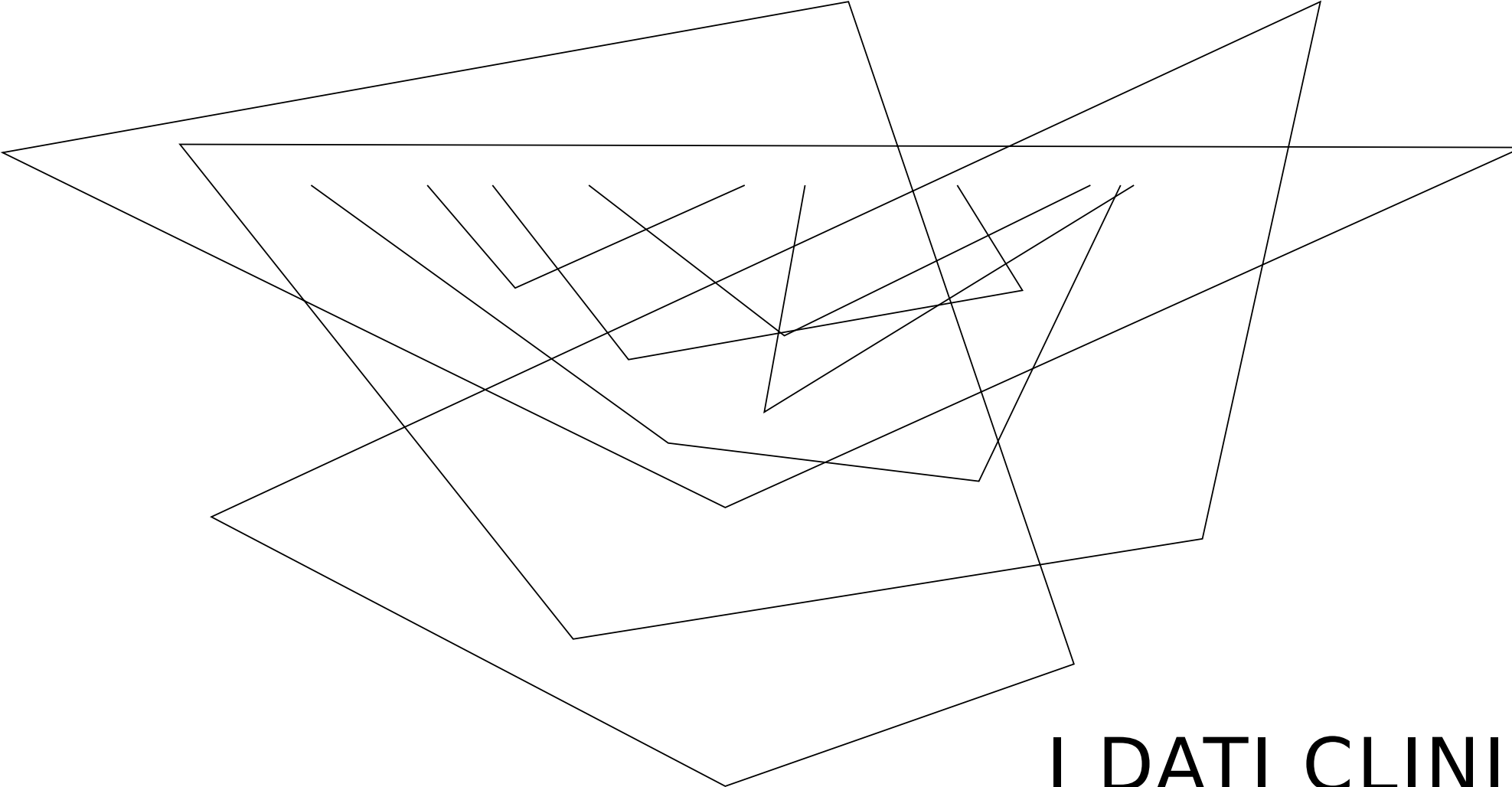
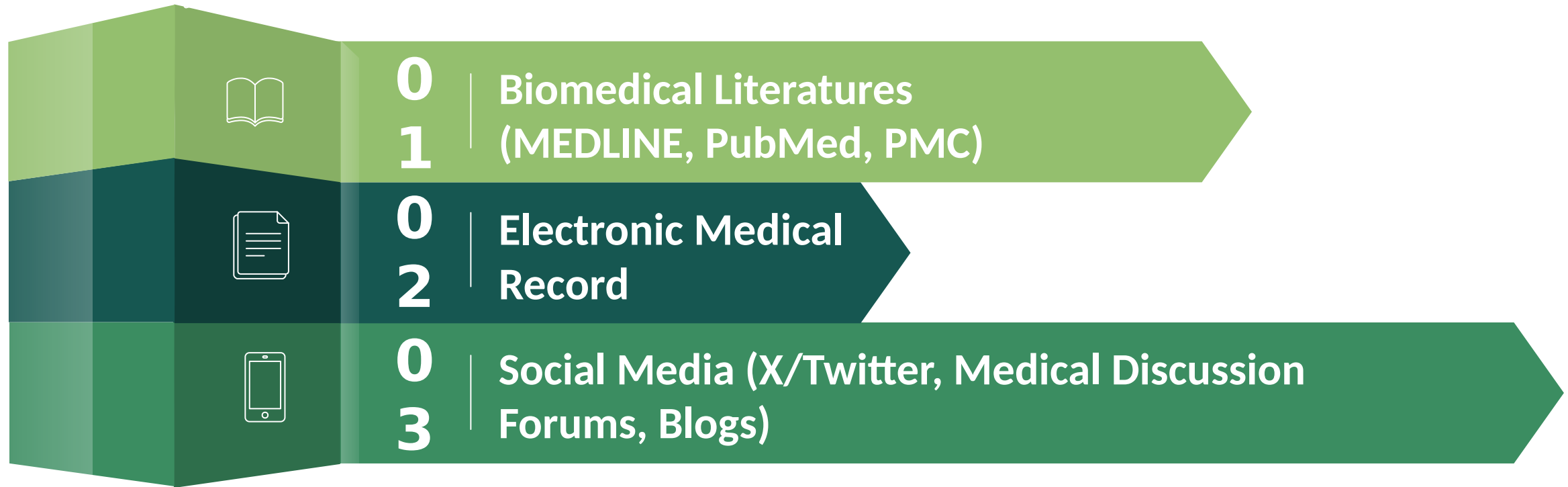




I DATI CLINICI E I LORO PROBLEMI

(again, thanks to Shweta Yadav)

Types Of Biomedical Data and its Sources



Biomedical Literature

MEDLINE



MEDLINE is a bibliographic database of life sciences and biomedical information. It includes bibliographic information for articles from academic journals covering medicine, nursing, pharmacy, dentistry, veterinary medicine, and health care. [Wikipedia](#)

History: 1879-present

No. of records: Over 26 million

Record depth: NLM Medical subject headings, abstracts, indexing

Cost: Free

Temporal coverage: 1946-present

Owner: [Lister Hill National Center for Biomedical Communications](#)

PubMed



Contact

Research center [United States National Library of Medicine \(NLM\)](#)

Release date [January 1996; 26 years ago](#)

Access

Website pubmed.ncbi.nlm.nih.gov 

Contains 33 million records

Task definition

Commentary

THE EMBO JOURNAL

The nature of the biological material and the irreproducibility problem in biomedical research

George V Papamokos^{1,2,3,4}

Biomedical research has a reproducibility problem since various crucial landmark papers could not be independently reproduced. While there are many causes related to statistical analysis, methodology or insufficient reporting of experimental details, this commentary argues that the complexity of biological material itself is, until now, a largely ignored source of irreproducibility. By discussing examples from evolutionary biology, intrinsically disordered proteins and current biomedical research, it contends that some results are irreproducible because we do not have the knowledge, the tools or the analytical ability to understand biological complexity and how it can give rise to different results. Instead of casting irreproducible research out as bad or sloppy science, they should serve as an inspiration for pioneering research not just to develop such tools but also to attempt to explore what lies beneath our current inability to deal with complexity.

The EMBO Journal (2019) e101011

Reproducibility is a central dogma of science as expressed by Karl Popper: "...for we have seen that non-reproducible single occurrences are of no significance to science". However, the life sciences do have an irreproducibility problem for published research articles: during the past years, evidence of irreproducible results has been accumulating, and various methodological causes have been identified (Eisner, 2018) and extensively reviewed (see Eisner, 2018 and references therein).

Yet, the role of the unique nature of biological material itself has not been discussed analytically for its potential role in the irreproducibility problem. I suggest that irreproducibility may also be a combined result of the complexity of the biological material and of our finite ability to deal with it, provided that all other factors are addressed (Eisner, 2018). Irreproducibility in carefully selected experiments can also be interpreted to the end that our current methods, theories and techniques are either applied ipsilaterally or are insufficient to analyse and describe functions, properties and unknown or missing parts of the biological material.

To support this argument, I will first present the perception of the science of biology by Ernst Mayr, one of the leading evolutionary biologists of the past century who described the evolutionary process as unpredictable or, at best, only statistically predictable (Mayr, 1961). Subsequent recent advancements either challenge the impossible or probabilistic approach of repeatability and predictability at certain levels, or suggest that predictable biochemical changes in evolutionary biology do not have a predictable molecular basis. A second argument is the discovery of a whole new kind of proteins, intrinsically disordered proteins (IDPs) (Dyson & Wright, 2005), that are key factors in health, disease, evolution and now possible drug targets. Lastly, two key examples at the level of eukaryotic cells (Hines-William *et al.*, 2014) and at the organismal level (Lithgow *et al.*, 2017) that were unable to account for different states adopted by biological material upon infinitesimal changes in the microenvironment which support that this is an opportunity to produce new science. I will also introduce methodology bias as a new term for the discussion of reproducibility.

Perception of biology by Ernst Mayr

Ernst Mayr, in his classic work "The Growth of Biological Thought", epitomized the philosophy of research in biology. He argued that the complexity of biological systems inevitably places biology beyond the limits of the natural sciences as the diversity of biological systems is based on rules that are substantially different from the laws of physics. Mayr believed that these rules cannot be described by methodological reductionism and can only be approached probabilistically, and that mathematics, although fundamental in physics, can contribute less to biology.

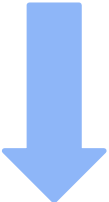
In his landmark paper *Cause and Effect in Biology* (Mayr, 1961), Mayr described biology as consisting of two different schools of thought: functional biology aims to answer the question "How?" while evolutionary biology attempts to answer "Why?" To this end, he used a warbler on his summer place in New Hampshire as an example to distinguish between proximate causes that are linked to functional biology and ultimate causes linked to evolutionary biology. Thus, the physiological condition of the bird—its sensing of photoperiodicity and temperature—is the proximate cause of migration, while the reduced availability of

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DOI: 10.15252/embo.j.2018101011

Unstructured Text

“Analysis of myeloid-associated genes in human hematopoietic progenitor cells.”

“Analysis of myeloid-associated genes in human hematopoietic progenitor cells.”



Entity Extraction

relation	subject	object
is_a	p53	protein
is_a	Bax	protein
has_function	p53	apoptosis
has_function	Bax	induction
involved_in	apoptosis	cell_death
is_in	Bax	cytoplasm
related_to	apoptosis	caspase_activation
...	...	...

Relation Extraction

structured knowledge: summary for a machine

Gene Mention Identification

- Focused on the identification of gene or protein names in biomedical text

*“Furthermore, as in the human gene, the 3' end of the **Cacna1f** maps within 5 kb of the 5' end of the **mouse synaptophysin gene** in a region orthologous to Xp11/23.”*

Gene Name

Gene Name

Gene Normalization

- **Gene normalization task aims to link the entity (gene) mentions in text to categories or concepts of a reference vocabulary.**
- **Accurate entity normalization enables us to summarize and compare biomedical insights across studies and obtain a holistic view of biomedical knowledge.**

Gene Normalization

Abstract (about *Drosophila melanogaster*, the fruit fly)

A locus has been found, an allele of which causes a modification of some allozymes of the enzyme [esterase 6](#) in *Drosophila melanogaster*. There are two alleles of this locus, one of which is dominant to the other and results in increased electrophoretic mobility of affected allozymes. The locus responsible has been mapped to 3-56.7 on the standard genetic map ([Est-6](#) is at 3-36.8). Of 13 other enzyme systems analyzed, only [leucine aminopeptidase](#) is affected by the modifier locus. Neuraminidase incubations of homogenates altered the electrophoretic mobility of [esterase 6](#) allozymes, but the mobility differences found are not large enough to conclude that [esterase 6](#) is sialylated.

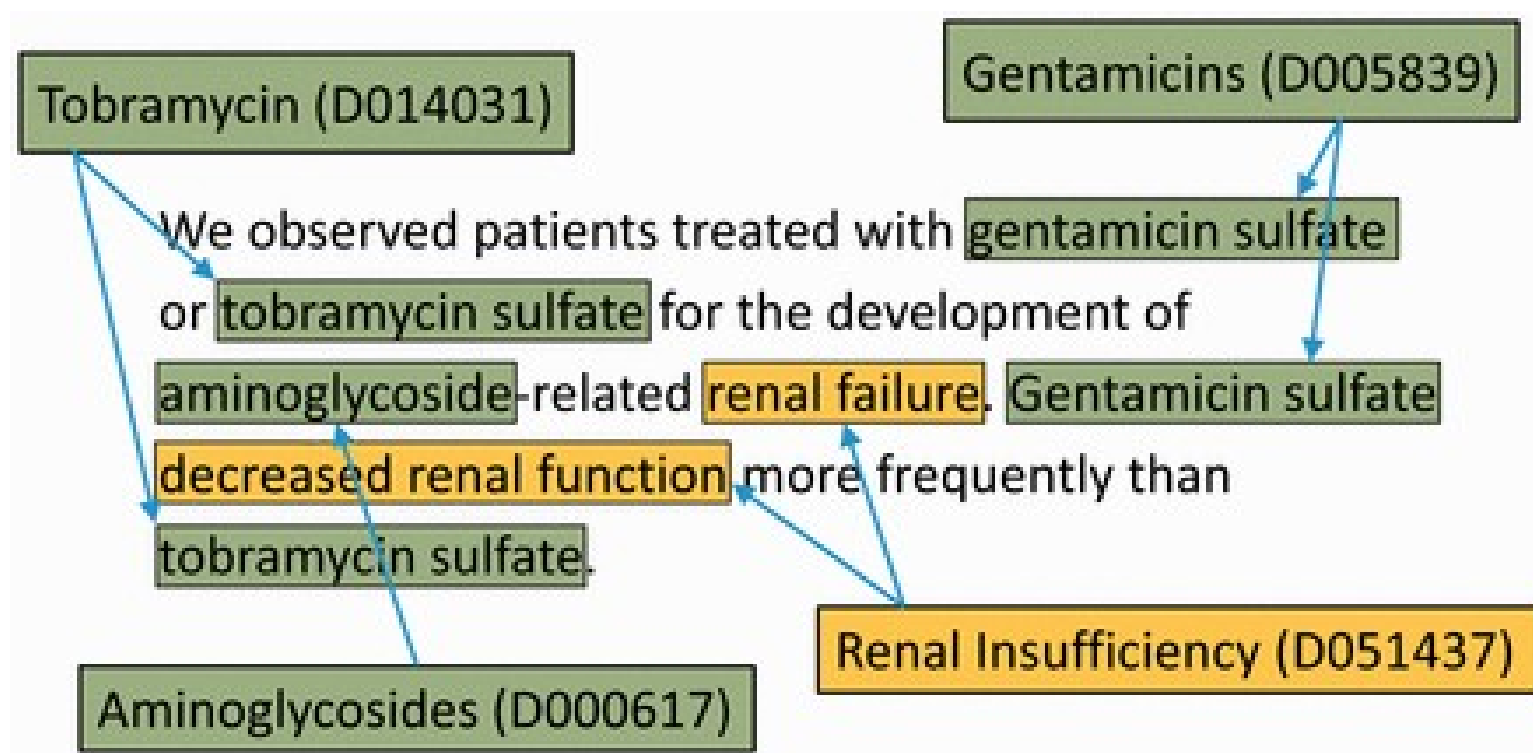
From FlyBase to NCBI Entrez Gene IDs

- FBgn0000592: [esterase 6](#), [Est-6](#)  38467 (*)
- FBgn0026412: [leucine aminopeptidase](#)  40636 (*)

(*) Disclaimer: specific mappings to Entrez Gene ID suggested by ChatGPT, not able to verify

Chemical Disease Recognition

Given a set of PubMed abstracts, return the normalized disease concept identifiers and the chemical entities mentioned in the document



Challenges

- Variability
- Pervasive ambiguity at every level of analysis

Challenges

"NACT is expressed in liver and brain in rat and shows preference for citrate over dicarboxylates...."

NACT:

- Neoadjuvant chemotherapy (PMID 8898170)
- N-acetyltransferase (PMID 10725313)
- Na⁺-coupled citrate transporter (PMID 12177002)

Challenges

mitogen activated protein kinase
kinase

CAT1 lacZ

3-fucosyl-N-acetyl-lactosamine

hairless sonic hedgehog

Challenges

- **Very Long Biomedical Named Entity**
 - “*human hematopoietic progenitor cells*”
- **Modifiers before basic Named Entity**
 - “*activated B cell lines*”
- **Complicated Constructions**
 - Two or more NEs share one head noun
 - *91 and 84 kDa proteins*

Challenges

- **Inconsistent naming convention**
- NF-kappaB, NFKappaB, NF-kappa B, (NF)-kappaB
- **Nested named entities**
- <Protein> <DNA> **Kappa 3** </DNA> **binding factor** </Protein>
- **Abbreviations of named entities**
- **TCEd, IFN, TPA**
- **Ambiguous NE**
- “**IL-2**” can be **protein** as well as **DNA** depending upon contexts

ELECTRONIC MEDICAL RECORDS

John B. Costello, M.D.

2821 North Ballas Rd Ste 165
St Louis, MO 63131

Phone: 314-995-9988
Fax: 314-995-7241

Wednesday, May 05, 2004

Patient

Mr. Steve A Pal
100 Anystreet
Saint Louis, MO 63146
44 year old Male
DOB: 01/01/60

Visit Date

Thursday, April 22, 2004

Chief Complaint

Sore Throat

History of Present Illness

Injection site identified and marked. Sterile pred with alcohol x3 and ethyl chloride.
LOCATION: throat *QUALITY:* scratchy *SEVERITY:* moderate *DURATION:* 2 days

Diagnosis - Major

Classic Heartburn 90901

Medications - Current

ACCUPRIL 20 mg 1 1/d Per Oral
Accolate 20 mg 1 2/d Per Oral
ACTONEL 30 mg 1 1/d Per Oral

Clinical note small excerpts

a/p:

This is 28 y/o G6P4013 presents for repeat caesarean delivery and bilateral tuba ligation. admit to labor and delivery cbc, type and screen iv fluids consented for repeat cesarean delivery, possible hysterectomy, possible blood transfusion, and bilateral tubal ligation. Hgb 11.1 patient alst ate at 2100on <DATE> discussed with the team <NAME> MD Obstetrics and Gynecology

Resident Attending note

I saw patient prior to scheduled repeat c/section this morning, reviewed cerner and counseled patient r/b/a. Patient with history of prior c/s x 1 here for RCD and BTL. Labs this morning includes Hgb 11.1, plat 207, bl gp O+ve. Singleton cephalic, posterior placenta. Writtent consent obtained, will proceed when OR is ready. Patient voiced understanding of the plan. <NAME>,MD

ADMISSION DATE: 03/24/97

DISCHARGE DATE: 03/28/97

The patient was **admitted** for her **gait disorder** ...

She was initially placed on **heparin** for **two days**, until her **MRI** results were obtained.

Information Extraction from Clinical Notes: i2b2 / n2c2 concept + relation annotation

5 CHIEF COMPLAINT: This 5-year-old male presents to Children's Hospital Emergency Department by the mother with "have **asthma**." Mother states he has been **wheezing** and **coughing**. They saw their primary medical doctor. He was evaluated at the clinic, given the **breathing treatment** and discharged home, was **not having** **asthma**, prescribed **prednisone** and an **antibiotic**. They told to go to the ER if he got worse. He has had **some vomiting** and **some abdominal pain**. His **peak flows** on the morning are normal at 150, but in the morning, they were down to 100 and subsequently decreased to 75 over the course of the day.

6 PAST MEDICAL HISTORY: **Asthma** with his last admission in 07/2007. Also inclusive of **frequent pneumonia** by report.

Informatics for Integrating Biology and the Bedside(i2b2)
(<http://www.i2b2.org>)

Chapman, W. W., Nadkarni, P. M., Hirschman, L., D'Avolio, L. W., Savova, G. K., & Uzuner, Ö. (2011). *Overcoming barriers to NLP for clinical text: the role of shared tasks and the need for additional creative solutions*. JAMIA, 18(5), 540–543.

Uzuner, Ö., South, B. R., Shen, S., & DuVall, S. L. (2011). *2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text*. JAMIA, 18(5), 552–556.

Caso clinico (E3C [Magnini et al 2021]) – da annotare per: malattia / sintomo / procedura / sito anatomico / farmaco

Anna A. e' una donna di 47 anni, vive con il figlio in un piccolo appartamento in un paese carsico, lavora nel campo della ristorazione e, nonostante una patologia genetica familiare che provoca delle alterazioni scheletriche, cardiologiche, oculari e cutanee, vive una vita serena e tranquilla. Ha una buona rete familiare e informale che supporta la famiglia. Viene seguita da un centro di riferimento specifico per patologia rara a Bologna e si sottopone a controlli clinicostrumentali periodici e regolari. Nel gennaio del 2017 esegue un intervento chirurgico di piede torto congenito presso l'Istituto Rizzoli. Durante la degenza, a seguito di un dolore retrosternale e di una sincope, viene sottoposta a un intervento di endoprotesi di aorta per dissezione acuta. Al rientro a casa, nel marzo del 2017, deve proseguire i controlli cardiologici ed eseguire la fisioterapia per gli esiti di intervento al piede e regolari controlli ematici. Non essendo autonoma nella deambulazione, viene presa in carico dai servizi del Distretto n. 1 dell'ASUGI di Trieste. Anna sembra, in un primo momento, una persona collaborante al piano assistenziale predisposto, ma, dopo un paio di interventi per controllare gli esiti della ferita al piede, si intuisce che qualcosa non funziona. Si somministra la scheda PHE-S® per una valutazione dell'elaborazione del vissuto della malattia, riscontrando un punteggio di 2,5. Parlando con lei emerge una profonda difficoltà ad accettare l'intervento al cuore, riferisce di sentirsi bloccata e ripete spesso: Sono in black out. ... [mezza pagina omessa] ...I valori della PHE-S dopo 4 mesi di interventi sono incrementati fino a raggiungere un punteggio pari a 3. ... [un'altra pagina]

Ma quanti sono questi dati? 7 neonati in intensive care (NICU)

Patient ID	Site	# MD notes	# RN notes	# PT notes	# OT notes	# SLP notes	# SW notes
1	UIC	78	86	4	10	4	9
2	UIC	404	542	15	57	4	4
3	MU	28	6	5	0	9	8
4	MU	321	36	31	30	12	29
5	MU	230	39	22	16	13	20
6	IOWA	272	17	11	6	6	20
7	IOWA	483	39	40	20	5	23

Major Challenge

Annotation Bottleneck

- Supervised Learning requires large-scale corpus
- Not Scalable for Expert to Annotate
- Crowdsourcing: Lack domain expertise

Digressione

(ma importante)

Intercoder /
interannotator
agreement

Intercoder agreement: Setting the stage ...

A: Andiamo al cinema stasera?
B: **Si'**, a che ora?

A: Andiamo al cinema stasera?
B: **Si'** vorrei, pero' ho da studiare

Two coders classify 150 occurrences of OK (Si')
into:

- Accept
- Acknowledgement (Ack)

Contingency Table

Coder 1

Coder 2

	Accept	Ack	Total
Accept	70	25	95
Ack	0	55	55
Total	70	80	150

Observed agreement: $125/150 = 83\%$

Random agreement: 50%

Intercoder agreement coefficients

- Cohen's Kappa (observed/expected agreement)

$$\kappa = (P(A) - P(E)) / (1 - P(E))$$

- Krippendorff's α (observed / expected disagreement)

$$\alpha = 1 - \frac{D_o}{D_e}$$

Cohen, J. (1960). *A coefficient of agreement for nominal scales*. Educational and Psychological Measurement.

Krippendorff, K. (1970). *Estimating the reliability, systematic error and random error of interval data*. Educational and psychological measurement.

The Computation of Kappa

$$\kappa = (P(A) - P(E)) / (1 - P(E))$$

- P(A): observed agreement (83%)
- P(E): expected agreement

Cohen, J. (1960). *A coefficient of agreement for nominal scales*.
Educational and Psychological Measurement.

Step 1: Proportions

	Accept	Ack	Total
Accept	70	25	95
Ack	0	55	55
Total	70	80	150

- $P(\text{Accept}, 1) = 95/150$
- $P(\text{Ack}, 1) = 55/150$
- $P(\text{Accept}, 2) = 70/150$
- $P(\text{Ack}, 2) = 80/150$

Step 2: Chance Agreement by item

- Accept: $(95/150) * (70/150) = 0.2956$
- Ack: $(55/150) * (80/150) = 0.1956$

Step 3: P(E) Expected Agreement

$$P(E) = 0.2956 + 0.1956 = 0.4912$$

$$\kappa = (P(A) - P(E)) / (1 - P(E))$$

$$\begin{aligned}\kappa &= (0.8333 - 0.4912) / (1 - 0.4912) \\ &= 0.6724\end{aligned}$$

$$\kappa = 0.6724 ???$$

Krippendorff (1980)

*Landis & Koch, 1977;
Rietveld & van Hout, 1993*

- $\kappa < 0.67$: bad
- $0.67 \leq \kappa < 0.80$: tentative
- $\kappa \geq 0.80$: definite
- $0 < \kappa \leq 0.20$: poor
- $0.20 < \kappa \leq 0.40$: fair
- $0.40 < \kappa \leq 0.60$: moderate
- $0.60 < \kappa \leq 0.80$: substantial
- $\kappa > 0.80$: almost perfect

Krippendorff, K. (1980). *Content analysis: an introduction to its methodology*. Beverly Hills, CA: Sage Publications.

Landis, J.R., & Koch, G.G. (1977). *The measurement of observer agreement for categorical data*. Biometrics, 33 (1), 159–174.

Rietveld, T., & van Hout, R. (1993). *Statistical techniques for the study of language and language behaviour*. Berlin - New York: Mouton de Gruyter.

Annotation con LLM

Prompt A.1: Prompt for Distillation

You are an advanced medical informatics specialist trained in understanding and aligning medical concepts. Your task is to analyze the following text from {profession} notes of {patient_type} patients, which may include contributions from MD, OT, PT, RN, and SLP, and extract every medical concept/entity that is relevant to the codes below. You are supposed to extract entities that are exactly present in the medical note.

##Note: Output the extracted entities in the following strict format without giving any justification and without any repetition:

###STRUCTURE BEGINS### code1_name: entity1, entity2, ... entityn

code2_name: entity1, entity2, ... entityn

....

code_n: entity1, entity2, ... entityn

###STRUCTURE ENDS###

with each entity separated by comma and result for each code in a separate line.

Answer empty string ' ' if you do not find any relevant entity belonging to the given code in the give medical note.

You are not supposed to give any justification and do not repeat the same entity. Strictly follow the structure given to you.

Do not treat dates and numerical values as entity.

You are supposed to provide entities based on the definition of the given definition of codes.

The code of interest and their respective definitions are given below:

##code_details##

The medical note is: ##medical_note##

Answer:

De-identification of Medical Records: 18 HIPAA identifiers



Mr. <XXX_Patient> is a <XXX_AGE> old white male with a history of diabetes mellitus who underwent a surgery on <XXX_DATE>. He was transferred to <XXX_Hospital> for endoscopy.

01

Raw Electronic Medical Record

Mr. John Smith is a 60 year old white male with a history of diabetes mellitus who underwent a surgery on November 15. He was transferred to Valwatnal Community Hospital for endoscopy.

De-Identified Medical Record.

02

Challenges

Medical Text

- **Variable Formatting**
 - **Inseparable phrase:**
“Neuro. nonfocal”
- **Structure without sentences**
 - **“Height: (in) 75 Weight (lb): 245 BSA (m2): 2.39 m2 BP (mm Hg): 92/52 HR (bpm): 120”**
 - **“Digoxin 250 mcg Tablet Sig: One (1) Tablet PO DAILY (Daily).”**
- **Missing punctuation**
 - **“no m/r/g b/l coarse rhonchi w/assoc upper airway sounds”**
 - **“Well appearing male in no acute distress Chest is clear No asthma”**

Challenges

Medical Text

- **Variety of textual subjects**
 - “I do not have an explanation for this dyspnea.”
 - **Diagnosis:** “He has - fever”
- **Language specific to medical context**
 - **Jargon:** “septic picture likely aspiration pneumonia secondary to dobhoff placement.”
 - **Ad-hoc abbreviations:** “His iron studies, LDH/Bili/Hapto, B12/Folate were normal.”
 - **Acronyms:** “recent tension PTX at OSH”

Challenges

Medical Text

- Misspellings
 - e.g. “Mitral stenosis is not present and definate [sic] mitral regurgitation is not seen.”
 - e.g. “s/p gsw now with fevers r/o abcess [sic]”
 - e.g. “Sclarea [sic] anicteric.”

SOCIAL MEDIA



verry



10:44 AM · Oct 2, 2020 · Twitter for iPhone

ignore my greasy eyebrows i just put castor oil on lmao but why is my eyelid like that ong it's swollen and lumpy.. i'm like 60% sure it's an eyelash glue **allergy** but i can't tell HDKDH



Hi all,

ive been suffering pretty bad the last few weeks with my [anxiety](#) and this week have felt very dizzy ans lightheaded. is this common for anxiety? Or is it possibly something else? does anyone feel this way when their anxiety is up?

0 likes, 9 replies

Information Extraction from Social Media Text

Tweet 1: I'm starting to think my **Paroxetine** turns **panic attacks** into **fat**.

Tweet 2: Any ideas on treating **depression** naturally **Paxil** is making me **gain unwanted pounds** and **water weight**...Thanks

Automatic extraction of medication names in tweets

The goal of this task is to **extract the spans that mention a medication or dietary supplement in tweets.**

Tweet Text	Begin	End	span	drug normalized
Only 3 Arnica Balms left...	8	19	Arnica Balms	arnica balm
@user sudafed that I'm not sure I'm comfortable taking it.	7	13	sudafed	sudafed
@user no my body hurts, they prescribed me hydros and moltrin	44	49	hydros	hydrocodone

Challenges

Social Media Text

- **Multiple Context:**
 - **Tweet 1:** “Advil cured my horrific pain, but made my stomach upset”
 - **Tweet 2:** “Advil cured my upset stomach but gave me a horrific pain”
- **Multiple word form:**
 - Short word-forms: “need to sleep 24/7”
 - Misspelled word forms: “fluoxetine, it just make me so tired”
 - Abbreviated words: “CIT for Citopram”
 - Slangs: “seroquel knocked me out”
 - Implicit sense: “hard time getting some Z’s”

Challenges

Social Media Text

- **Usage of the figurative language**
 - “My skin is paper, razor is the pen”
- **Usage of implicit sense**
 - "You have a killer look"

Conversazioni fra pazienti e providers

Patient: Yeah, I don't, I don't do the frozen meal.

Educator: Okay.

Patient: I was basically doing the uh, vegetables.

Educator: Okay.

Patient: Frozen vegetables

Educator: They should be fine.

Patient: Yeah.

Educator: But... but, but I do want you to start looking at those nutrition labels.

Patient: Okay.

Educator: And look for something that says less than 5%.

Patient: Okay.



Cosa pensano i pazienti?

Interviews with hospitalized heart failure patients about their perspective on their illness

Patient: Quality of life is more valuable to me than working on the next project

Interviewer: Alright. Okay. So self care...

Patient: matter

Interviewer: Is primary. Yeah it matters. Yeah definitely.

Patient: Yeah

Interviewer: Now do you have some support at home? You girlfriend's around? She is going to help you out or?

Patient: Yeah. You know I am probably the most fortunate guy you are gonna meet today. I am fortunate in that even aside from my family, there are friends that genuinely care about me.

Interviewer: Alright

Patient: I mean really genuinely..genuinely care about me



Challenges

Patient

Conversations

- **Dinamica del parlato**
 - Turn taking
 - Overlap / interruptions
 - Particles
- **Usage of figurative language**
 - “My skin is paper, razor is the pen”
- **Usage of implicit sense**
 - “You have a killer look”

Shared Tasks

- BioCreative: Critical Assessment of Information Extraction in Biology
- National NLP Clinical Challenges (n2c2)
- BioNLP Workshop
- ClinicalNLP Workshop
- Text Retrieval Conference (TREC)